

Lesson 13 — Compound Conditions (AND / OR / NOT)

Matches LEGO Teacher Portal lesson: "Ocean Walker".

Objective (I can...): I can combine conditions with AND, OR, and NOT so my program reasons over more than one fact. **CS concept:** Compound boolean conditions; logical operators · **Builds on:** Lesson 12 (if/then/else).

AI Connection (2 min): A security system might act only if motion AND an unrecognized face are both detected. *First-principles question:* "When would requiring AND make a system safer, and when would it make it miss something important?"

Materials: LEGO CS & AI kit + device per pair, two kinds of triggers (object + color), a truth-table handout, notebooks. **Model reuse:** Conditionals model from Lesson 11 — code only.

60-minute flow - Warm-up (8): "True/false math" — pairs fill a small truth table for AND and OR using everyday statements. Introduce NOT as flipping true/false. - **Concept + demo (10):** Demo a compound condition: if (object near) AND (red seen) then act; change AND to OR and show how the behavior widens. Show NOT on a single condition. - **Build/Code (30):** Pairs code one AND condition and one OR condition driving different actions, then predict and test the result for each combination of inputs, logging a mini truth table in the notebook. *Swap at 15 min.* - **Challenge (7):** Add a NOT so an action fires only when a condition is *false*; verify it. - **Share + pack-up (5):** Pairs demo an AND vs. an OR case and explain the difference in plain English. Model stays built.

Pair roles: Driver codes, Navigator owns the truth table and calls each test case. Swap at 15 min. (*Trio: third runs the combinations.*)

Differentiation: *Support* — provide the truth table pre-filled; students match it to one compound condition. *Extension / fast-finisher* — build a three-condition expression and predict its table. *Reteach* — test each simple condition alone, then combine two and watch the result change.

Exit check: Pair's program uses at least one compound condition correctly and they can explain AND vs. OR. **At-home (optional):** Write a real rule that needs AND and one that needs OR (e.g., "wear a coat if cold AND going outside"). ~10 min.